Copilot Studio + ServiceNow

How the Pieces Fit Together

A plain-language guide to the solution architecture

“AI chatbot meets live human support — seamlessly.”

What Does This System Do?

01

AI Assistant

- Copilot Studio answers everyday questions
- Available instantly, 24/7
- Knows when to call a human

02

The Bridge

- Connects Microsoft and ServiceNow
- Tracks the chat state in real time
- Routes messages to the right place

03

Human CSR

- Works in normal ServiceNow workspace
- User sees their reply instantly
- No new tools for the rep to learn

The user stays in the **same chat window** the whole time — no switching apps, no hold music, no starting over.

Three Main Parts

Microsoft Side

Copilot Studio — the AI that chats with users

Front End — the chat window (browser or Teams)

Direct Line — connects the front end to the AI

Runs in the Microsoft cloud

The Bridge

Session tracker — knows if you're in bot or live mode

Message router — sends messages to the right place

WebSocket server — pushes rep replies to you instantly

Runs on your own server (container / VM)

ServiceNow Side

AWA Queue — lines up chats for the next available rep

CSR Workspace — where reps read and reply to chats

2 APIs + 1 Rule — the only custom setup required

Runs on your ServiceNow instance

Contoso Connect — Employee

pbqgkr6d.use.devtunnels.ms:5001

Set Google Chrome as your default browser and pin it to your taskbar | Set as default

Contoso Connect

Home | HR | IT Support | Benefits | Directory | Learning | Search

Good afternoon, Aria 👋

You have **3 open IT tickets**, the next all-hands starts in **2 days**, and 14 new teammates joined this week.

Ask the assistant

OPEN TICKETS	PTO BALANCE	PULSE SCORE	OFFICE CAPACITY
3	12d	8.4	62%
+1 since Monday	accrued +1d	+0.2 vs Q1	today

Latest announcements

View all →

Q2 all-hands — Friday at 11:00

Posted 2 hours ago • Leadership

Calendar invites are out. We'll cover Q1 results, the new product strategy, and a Q&A with the leadership team. See the agenda on the right →

VPN maintenance — Sunday 02:00–04:00 UTC

Posted yesterday • IT Operations

Brief disconnects expected during the window. No action required from you, but please save your work before EOD Saturday if you'll be on PTO and connecting from abroad.

Welcome our 14 newest teammates

Posted Monday • People & Culture

Engineering, sales, and ops grew this week. Say hi in #welcome and chec a few of them in the New Hires panel.

Log in | ServiceNow

https://dev379803.service-now.com/navpage.do

Chat

servicenow

User name

alex.rep

Password

Log in

Forgot Password ?

34°F Sunny | Search | 7:25 AM 4/25/2026

Copilot | Copilot | Microsoft Teams

https://teams.cloud.microsoft

Search (Ctrl+Alt+E)

Activity
Chat
Calendar
Calls
OneDrive
Copilot
Dynamics C...
Apps

Chat

Unread | Channels | Chats

- Copilot
- Quick views
  - Followed threads
  - Drafts
- Favorites
- Chats
  - IT Helper
  - OMS Returns Agent
  - SNOW MCP Agent
- Teams and channels
  - Contoso
    - General
    - Microsoft Helpdesk
  - CEO
    - Weekly Updates

See all your teams

IT Helper
Chat

8:54 AM

How do I connect to VPN?

IT Helper 8:54 AM

To connect to the Contoso VPN, open your VPN client and enter your Contoso credentials. If you need the specific server address or step-by-step instructions, please check your onboarding documentation or contact IT for your department's VPN setup. If you have trouble connecting, let me know and I can help further.

8:55 AM

How do I reset my password?

IT Helper 8:55 AM

You can reset your password by clicking the "Forgot Password" link on the login page, entering your email, and following the instructions sent to your inbox. You may also answer your security questions if prompted, or contact IT support for further assistance[1]. Please remember to enroll in self-service password reset at https://aka.ms/sspr for easier future resets.

Type a message

Inbox | Service Operations Worksp

https://dev379803.service-now.com/now/sow/inbox

servicenow | All | Service Operations ...

Inbox

Status
Available

No inbox items

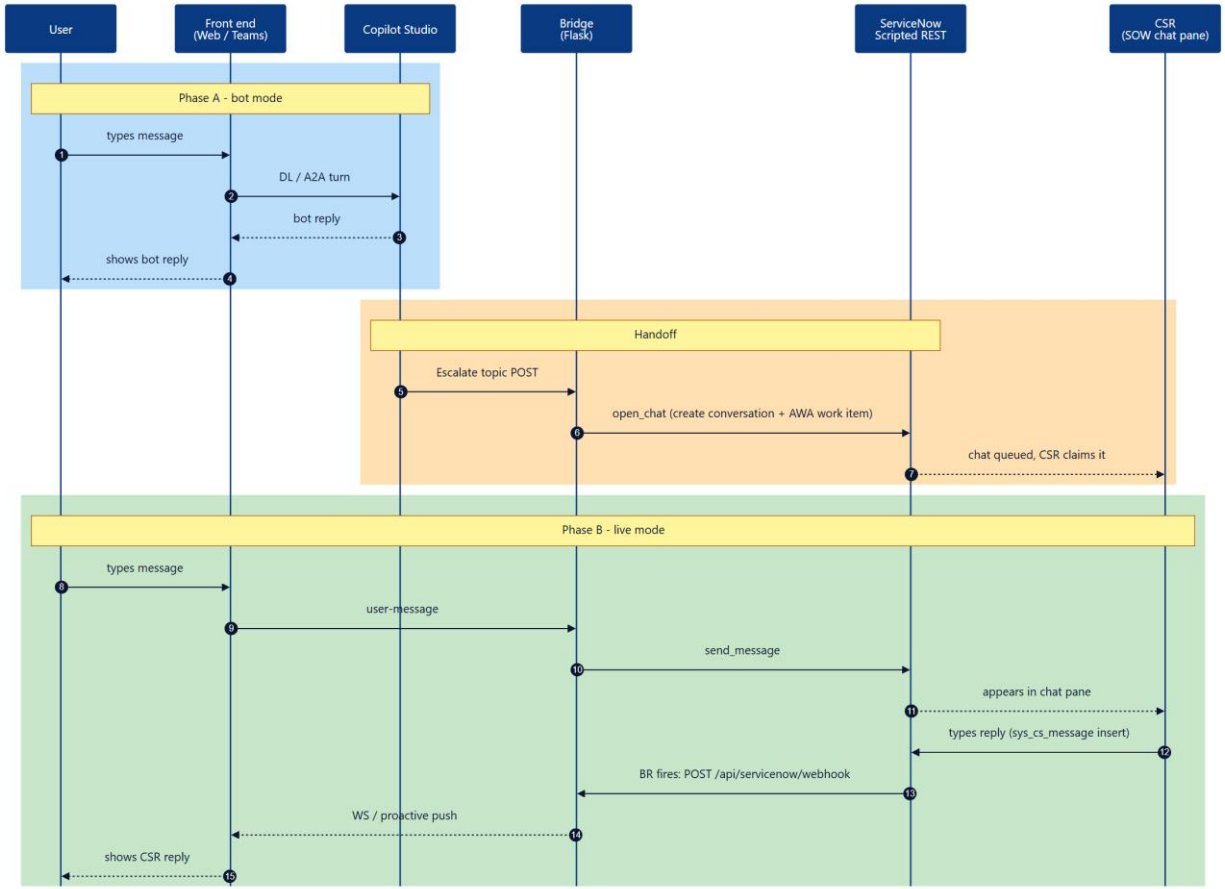
As work items are assigned, they will appear in this panel.

Check your Inbox

Click a card in the Inbox to open a work item in this workspace.

40°F Sunny | Search | 8:58 AM 4/25/2026

How a Conversation Flows — Start to Finish



The Bridge: The Glue in the Middle

What it does

- Runs on your own server (container or VM)
- Receives the “escalate” signal from Copilot Studio
- Calls ServiceNow to open a live-chat conversation
- Forwards user messages to the human rep
- Receives the rep’s replies via webhook
- Pushes replies to the user’s screen in real time

Chat States (what the user sees)

1. **BOT** — Chatting with the AI assistant. Normal bot replies.
2. **QUEUED** — "Connecting an agent…" — waiting for a human rep to claim the chat.
3. **LIVE** — Talking to a real human rep. Messages go directly to ServiceNow.
4. **CLOSED** — Chat ended. User can start a new session at any time.

Three Ways to Chat

Option A

Web Browser

- User opens a webpage with a chat widget
- Copilot Studio Web Chat handles the AI side
- Bridge handles the handoff to a human rep

Best for: intranet portals and self-service sites

Option B

Teams — Bot Mode

- The Teams app is the bot the user talks to
- Copilot Studio runs quietly in the background
- Rep replies are pushed into the Teams chat

Best for: Teams-first companies

Option C

Teams — CS Mode

- User chats with Copilot Studio in Teams
- When escalation is needed, our agent is called automatically
- Replies appear under the same AI assistant name

Best for: orgs with an existing Copilot Studio agent

All three channels use the same Bridge and the same ServiceNow setup.

What Gets Set Up in ServiceNow?

Only 3 custom pieces are needed — no extra apps or plugins.

1

open_chat API

When escalation happens, the Bridge calls this API in ServiceNow. It creates a real live-chat conversation and puts it in the queue so the next available rep can pick it up.

2

send_message API

Every message the user types (while connected to a human rep) is delivered to the rep through this API. The rep sees the messages appearing live in their workspace.

3

Outbound Business Rule

Every time a rep types a reply, ServiceNow runs this tiny script automatically. It sends that reply to the Bridge, which pushes it to the user's screen. Without this, rep replies would never leave ServiceNow.

That's it. No ServiceNow store apps. No extra plugins. Just 2 REST APIs and 1 Business Rule.

How Is It Kept Secure?

From	To	How it's protected	Production tip
Copilot Studio	Bridge	X-Agent-Secret header	*Rotate this key on a schedule*
Browser	Bridge	Unique session ID (server-issued)	*Never cache IDs from previous sessions*
ServiceNow	Bridge	X-Bridge-Secret header	*Stored as a secure sys_property in SN*
Bridge	ServiceNow	Dedicated service account + password	*Use a strong password; rotate regularly*

For production, also:

Use HTTPS only (no plain HTTP) • Add rate limiting on the escalate endpoint • Store bridge sessions in Redis (not in memory)

The Key Takeaways

One bridge, one ServiceNow setup — works for all three front-end channels (web, Teams bot, Teams CS mode).

The user stays in one seamless chat window. They never have to switch apps or start over.

The human rep works in their normal ServiceNow workspace — no new tools to learn.

ServiceNow needs minimal customization: 2 REST APIs and 1 Business Rule. No store apps or plugins.

The Bridge is the only stateful piece. For production, store sessions in Redis instead of memory.

Want to Go Deeper?

ServiceNow Setup Guide

02-servicenow-setup.md

Bridge Backend Reference

03-bridge-backend.md

Web Chat Setup

05-browser-webchat.md

Teams Channel Setup

13 & 14 teams setup docs

End-to-End Testing Guide

06-end-to-end-test.md

Production Hardening Checklist

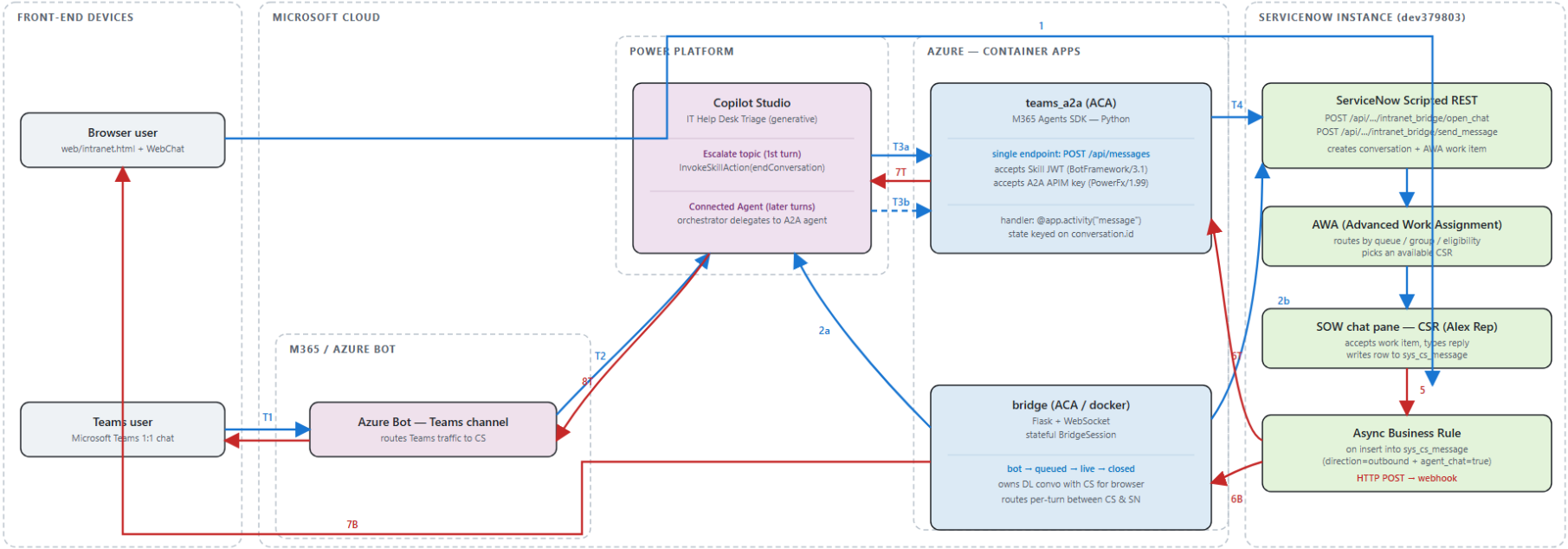
09-production-hardening.md

amacey-msft/copilot-studio-servicenow-bridge: Copilot Studio <-> ServiceNow live agent handoff bridge (CSM/AWA, Direct Line, Flask). Verified end-to-end reference with Scripted REST + Business Rule.

Copilot Studio — ServiceNow live-agent handoff

Two front-end paths (browser, Teams). Both end in the same ServiceNow conversation.

User → CSR (solid = first turn / Skill) — User → CSR (dashed = later turns / Connected Agent A2A) — CSR → User



Step legend

User → CSR

1. Browser → bridge (HTTPS / WS, bridge owns the Direct Line conversation with CS)
2a. bridge → CS (normal browser turn) 2b. bridge → SN REST (escalated browser turn)
T1. Teams user → Azure Bot T2. Azure Bot → CS
T3a. CS → teams_a2a via Skill (1st turn, InvokeSkillAction, Bearer JWT, BotFramework/3.1)
T3b. CS → teams_a2a via Connected Agent A2A (later turns, x-api-key, PowerFx/1.99-local)
T4. teams_a2a → SN Scripted REST (open_chat / send_message)

CSR → User

5. CSR types reply in SOW → row inserted into sys_cs_message → async BR fires
6T. BR → teams_a2a POST /api/sn-webhook 6B. BR → bridge POST /api/servicenow/webhook
7T. teams_a2a proactively pushes message to CS-signed service_url 7B. bridge pushes WS frame to browser
8T. CS reply → Azure Bot → Teams user

Where the M365 Agents SDK fits

teams_a2a is ONE ACA app, registered in CS in TWO places:
Settings → Skills ("ServiceNow Live-Agent Handoff")
Agents → Connected Agents ("ServiceNow Live Agent")
Both point at the same /api/messages endpoint. The SDK's @app.activity("message") handler treats both protocols identically; per-conversation handoff state is keyed on conversation.id.

Why the browser path skips teams_a2a

Browser is not a CS-native channel. The bridge owns its own Direct Line conversation with CS, so it plays the same router role for browser that teams_a2a plays for Teams.